Downloaded from gsabulletin.gsapubs.org on March 6, 2010

Geological Society of America Bulletin

Constraints on the motions of South American and African Shields during the Proterozoic: I. $^{40}Ar/^{39}Ar$ and paleomagnetic correlations between Venezuela and Liberia

T. C. ONSTOTT, R. B. HARGRAVES, DEREK YORK and CHRIS HALL

Geological Society of America Bulletin 1984;95;1045-1054
doi: 10.1130/0016-7606(1984)95<1045:COTMOS>2.0.CO;2

Email alerting services click www.gsapubs.org/cgi/alerts to receive free e-mail alerts when new articles cite this article

Subscribe click www.gsapubs.org/subscriptions/ to subscribe to Geological Society of America Bulletin

Permission request click http://www.geosociety.org/pubs/copyrt.htm#gsa to contact GSA

Copyright not claimed on content prepared wholly by U.S. government employees within scope of their employment. Individual scientists are hereby granted permission, without fees or further requests to GSA, to use a single figure, a single table, and/or a brief paragraph of text in subsequent works and to make unlimited copies of items in GSA's journals for noncommercial use in classrooms to further education and science. This file may not be posted to any Web site, but authors may post the abstracts only of their articles on their own or their organization's Web site providing the posting includes a reference to the article's full citation. GSA provides this and other forums for the presentation of diverse opinions and positions by scientists worldwide, regardless of their race, citizenship, gender, religion, or political viewpoint. Opinions presented in this publication do not reflect official positions of the Society.

Notes

Geological Society of America

THE GEOLOGICAL SOCIETY OF AMERICA

Downloaded from gsabulletin.gsapubs.org on March 6, 2010

Constraints on the motions of South American and African Shields during the Proterozoic: I. $^{40}Ar/^{39}Ar$ and paleomagnetic correlations between Venezuela and Liberia

T. C. ONSTOTT
R. B. HARGRAVES
Princeton University, Princeton, New Jersey 08544
DEREK YORK
CHRIS HALL
University of Toronto, Toronto, Ontario, Canada M5S 1A7

ABSTRACT

The Encrucijada pluton of Venezuela and amphibolites near Harper, Liberia, are located on opposite sides of the Liberian Pan-African mobile belt, when South America is restored in a fit described in the literature with respect to Africa. Both units yield stable, bipolar, high-temperature magnetizations that, on the basis of thermal demagnetization data, $^{40}Ar/^{39}Ar$ hornblende and biotite radiometric results, and Rb-Sr whole-rock and biotite radiometric results, appear to be indistinguishable in age at 1.9 to 2.0 Ga. With South America in this reconstruction, the corresponding paleomagnetic poles suggest that ~1,000 km of right-lateral strike-slip motion has occurred between the West African and Guayana Shields, probably along the Liberian Pan-African Belt. Furthermore, the Encrucijada and Harper poles are distinct from the 1.9- to 2.0-Ga paleomagnetic poles from the Kalahari Shield and tentatively suggest that relative motion has occurred between the Kalahari Shield and the West African and Guayana Shields since that time.

INTRODUCTION

Hurley and Rand (1969) showed that the Proterozoic Eburnean and the Archean Liberian age provinces of West Africa correlate with the Proterozoic and Archean age provinces of the Guayana Shield of South America (Fig. 1). On the basis of their implied proximity since the Archean, Hurley (1973) proposed that the transecting 0.5-Ga Pan-African mobile belt in Liberia (see Fig. 1) developed without major motions between the shields.

In a Bullard and others (1965) fit, however, the boundaries separating the Archean and Proterozoic age provinces (GFZ and DFZ in Fig. 1) and the paleopoles are offset by ~1,000 km, as if there was right-lateral motion between the West African and Guayana Shields, along the Pan-African mobile belt (Onstott and Hargraves, 1981). The paleopole offset could not be demonstrated conclusively, however, because the "magnetization ages" were poorly constrained. Sufficiently precise ages of magnetization can be obtained by using $^{40}Ar/^{39}Ar$ analyses of hornblendes and biotites, as has been shown by Berger and York (1979a, 1979b, 1981a) for metamorphic rocks of the Canadian Shield.

In this study, we evaluate the tectonic hypotheses of Onstott and Hargraves (1981) by a combined $^{40}Ar/^{39}Ar$ and paleomagnetic study of two localities on opposite sides of the Liberian Pan-African mobile belt, the La Encrucijada granite pluton of Venezuela and the amphibolites near Harper, Liberia (1 and 2 in Fig. 1). Our findings are presented in two parts. In this paper, Part I, we compare the high-temperature history and magnetizations of these two localities. We then synthesize these data with published paleomagnetic data from South Africa in order to constrain the relative motion of the Kalahari Shield with respect to the Guayana and West African Shields. In Part II (the second paper), we present cooling histories for the Venezuelan and Liberian localities based on the $^{40}Ar/^{39}Ar$ thermochronometry method of Berger and York (1981b) and compare the age spectra with theoretical cooling models. We then use these cooling histories to estimate the ages of the secondary magnetizations identified in the paleomagnetic study and to compare the $^{40}Ar/^{39}Ar$ thermochronometry with the magnetothermometry.

GENERAL GEOLOGY

Our study focused on the granite pluton at La Encrucijada, Venezuela, and the amphibolites near Harper, Liberia, because published radiometric data, reviewed below, suggested that their cooling histories since peak metamorphism, and thus their "magnetization" ages, might be quite similar. The rock types and geologic setting of these two localities, however, are very different.

The Encrucijada granite pluton is located on the northern edge of the Guayana Shield (see Fig. 2). It is ~10 km in diameter and is composed mainly of slightly gneissic quartz monzonite that contains minor proportions of biotite, hornblende, and sphene (see Appendix 1)[1] and is commonly medium- to coarse-grained but locally porphyritic (Posadas and Kalliokoski, 1967). It intrudes the quartzofeldspathic gneisses of the Imataca Complex, which locally are in the almandine-amphibolite facies of regional metamorphism (Kalliokoski, 1965). Posadas and Kalliokoski (1967) reported a Rb-Sr whole-rock isochron age of 2,064 ± 87 Ma (1 S.E.) (initial $^{87}Sr/^{86}Sr$ = 0.7024 ± 0.0017) for the pluton. Short and Steenken (1962) reported a Rb-Sr age of 1,958 ± 45 Ma (1 S.E.) for biotite from the pluton. Both Rb-Sr dates have been recalculated with the decay and isotopic constants of Steiger and Jäger (1977). These ages coincide with the timing of the Trans-Amazonian Orogeny and suggest the pluton was intruded from a low initial strontium source

[1]Appendices 1 through 4 may be secured free of charge by requesting Supplementary Data 84-18 from the GSA Documents Secretary.

Additional material for this article, Appendices 1 through 4, may be obtained free of charge by requesting Supplementary Data 84-18 from the GSA Documents Secretary.

Geological Society of America Bulletin, v. 95, p. 1045–1054, 9 figs., 2 tables, September 1984.

Downloaded from gsabulletin.gsapubs.org on March 6, 2010

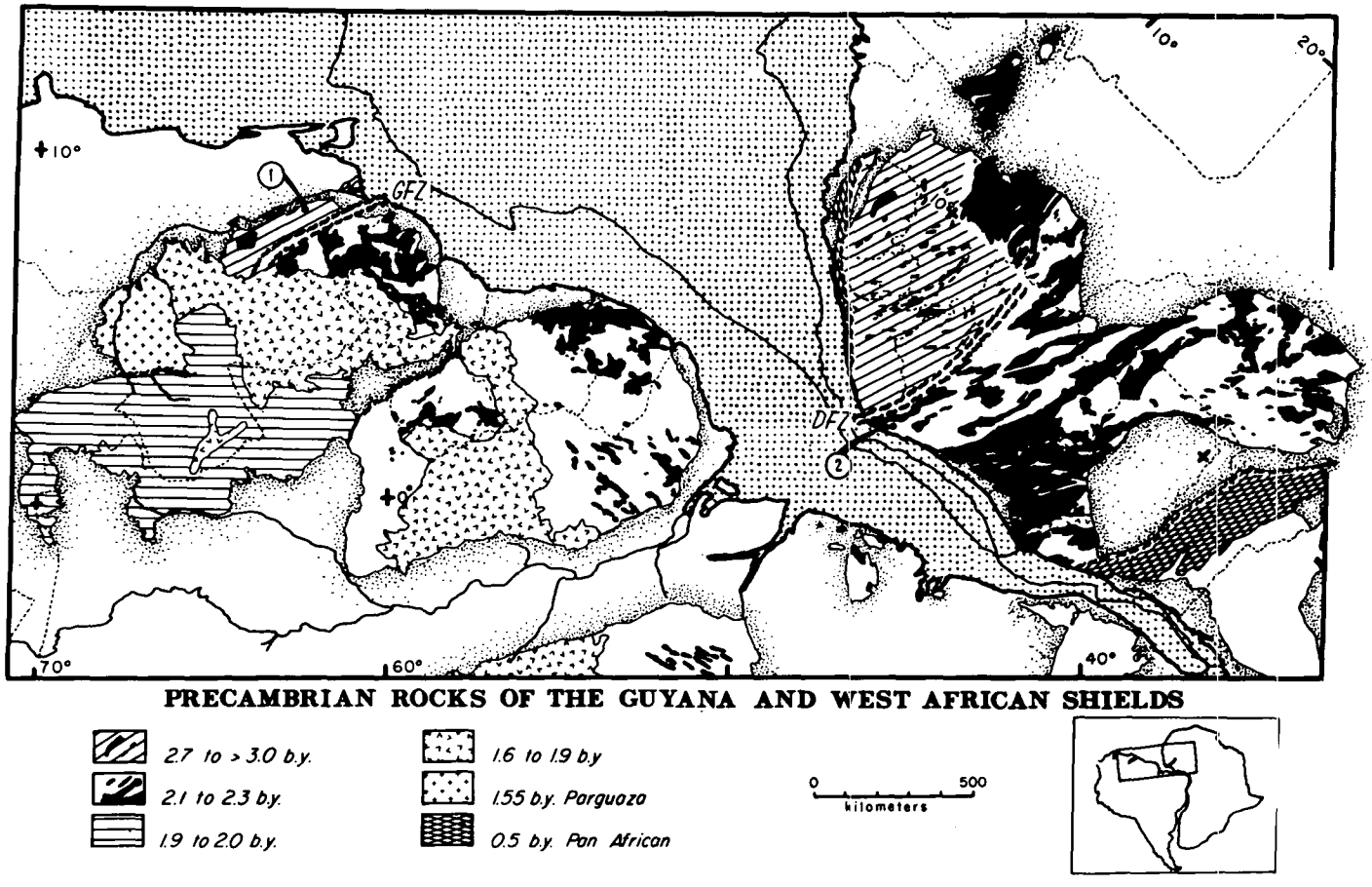


Figure 1. West Africa and South America shown positioned according to the reconstruction of Bullard and others (1965). The major Archean and Proterozoic units are shown (adapted from a map by Townsend and Gibbs, Pl. 4 of Gibbs, 1975). GFZ is the Guri fault zone, and DFZ is the Dube fault zone. Location 1 is the pluton near Encrucijada, Venezuela, and location 2 represents the amphibolite near Harper, Liberia.

region into the Imataca Complex close to the peak of the orogeny (Montgomery and others, 1977).

The amphibolites that outcrop in the vicinity of Harper in Liberia are located on the southern margin of the West African Shield (see Fig. 3). They range from medium to coarse grained and contain varying proportions of plagioclase, hornblende, augite, garnet, and sphene (see Appendix 1). The presence of hornblende-garnet-sillimanite-plagioclase assemblages nearby indicates that the peak metamorphic facies was upper amphibolite. Tight, isoclinal folds and anatectic textures predominate (Brock and others, 1977). Hedge and others (1975) published a K-Ar biotite age of 1,830 ± 40 Ma (2 S.E.) and 2 K-Ar hornblende ages of 1,970 ± 60 and 1,820 ± 50 Ma (2 S.E.) from this vicinity. Rb-Sr whole-rock isotopic analyses for quartzofeldspathic gneisses that envelop the amphibolites in this region (Hurley and others, 1971) yield an isochron age of 1,983 ± 34 Ma (1 S.E.) (initial $^{87}Sr/^{86}Sr = 0.7097$) when fitted with the York regression method (York, 1969). The amphibolites evidently represent higher-grade equivalents of 2.2-Ga Birrimian supracrustals of Ivory Coast and Ghana (Cahen, 1961; Kolbe and others, 1967) but underwent amphibolite-facies metamorphism during the 2.0-Ga Eburnean Orogeny.

FIELD AND LABORATORY PROCEDURES

Oriented samples (37) were collected from fresh road-cuts along a highway traversing the Encrucijada pluton (see Fig. 2), and 25 oriented samples were collected from fresh road-cuts, quarries, and beach exposures in the amphibolites near Harper (see Fig. 3).

Specimens were stepwise AF and thermally demagnetized (for further details see Appendix 2), and the data were analyzed by use of orthogonal plots (Zijderveld, 1967). Samples typically yielded two to four components of magnetization. Only 1 component generally remained after AF demagnetization at 20 to 30 mT, beyond which the remanence either followed a straight-line trajectory toward the origin or formed a "stable end point" cluster of vectors. Above 60 to 70 mT, the remanence generally began to scatter. The least-squares orientation of the linear trajectories and stable end points within this 25 to 60 mT "window" were calculated using the method proposed by Kirschvink (1980). Analyses of the less stable "secondary" components are discussed in the second paper. The mean directions and Fisherian statistics (Fisher, 1953) were calculated by averaging at the sample level only those least-squares directions composed of four or more consecutive demagnetization steps and yielding a maximum angular deviation (MAD) of $<20°$. The results are summarized in Table 1. The detailed demagnetization data are presented in Appendix 3.

High-purity separates of hornblende, biotite, sphene, and perthite were removed from one sample of the Encrucijada pluton for $^{40}Ar/^{39}Ar$ analyses. Similarly, high-purity separates of hornblende, biotite, and plagioclase were removed from one sample of the Harper amphibolite.

Procedures used for sample irradiation and

Downloaded from gsabulletin.gsapubs.org on March 6, 2010

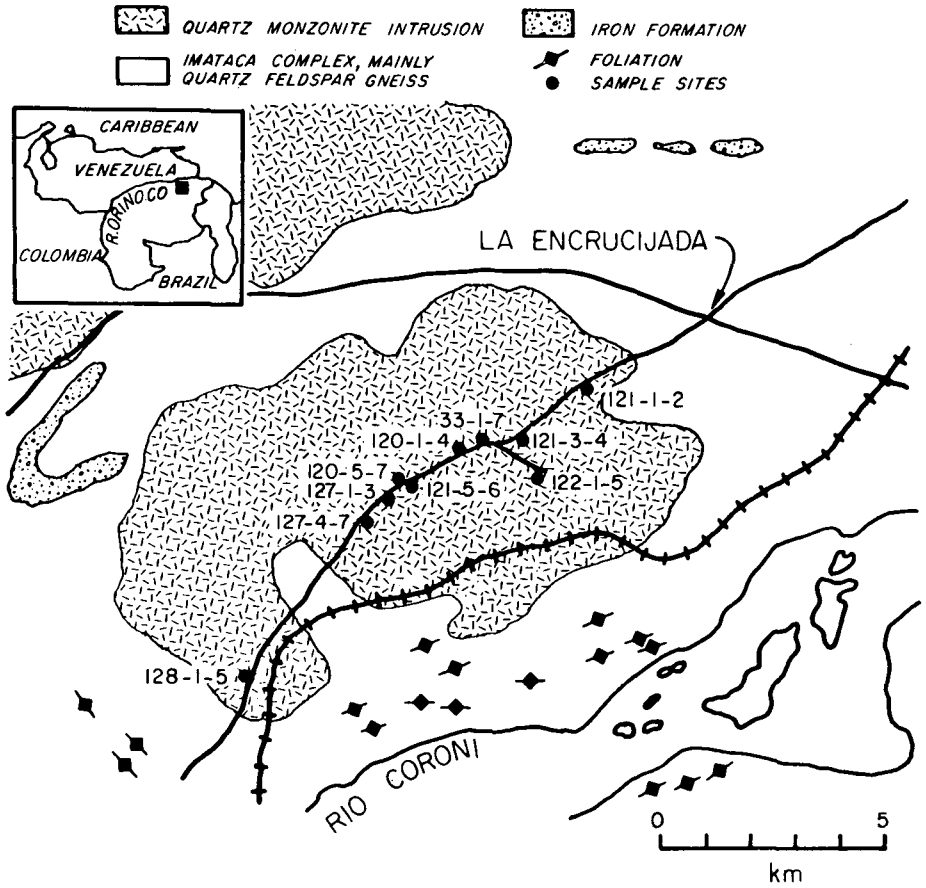


Figure 2. General geology and sampling localities for the Encrucijada quartz monzonite pluton southwest of Puerto Ordaz, Venezuela (adapted from Fig. 1 of Posadas and Kalliokoski, 1967).

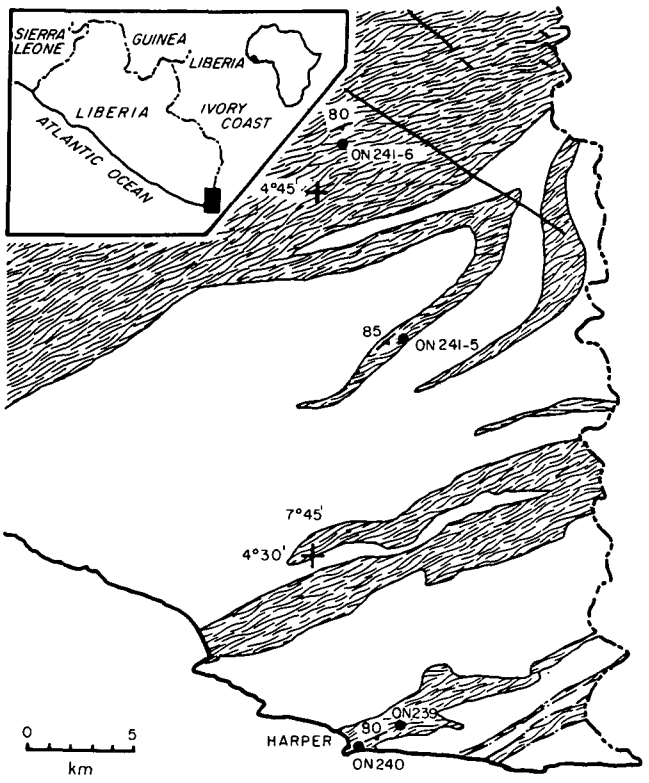


Figure 3. General geology and sampling localities for the amphibolites in the schist belts near Harper, Liberia (adapted from the geologic map of Harper Quadrangle, Liberia: Map I 780-D by Brock and others, 1977).

argon extraction are similar to those of Berger and York (1981b) (see Appendix 4 for details). In the calculation of ages, we used the correction factors for interference isotopes listed in Masliwec (1981), the error propagation equations derived in Hall (1982), and the decay and isotopic constants of Steiger and Jäger (1977).

For any sample for which the ages of 4 or more consecutive temperature steps agree within 2-sigma error limits, a plateau age was determined by calculating an integrated age for that portion of the age spectrum on the plateau. The detailed isotopic data for the hornblendes and biotites are presented in Table 2. Analyses of the feldspars and sphene are presented in Part II.

RESULTS

Argon Ages

Comparison of the spectra for hornblendes from the Harper amphibolite and the Encrucijada pluton reveals that the ages of the high-temperature portions of their spectra corresponding to >60% release of $^{39}Ar_K$ overlap within 1 sigma error (see Fig. 4). The Harper hornblende yields a plateau age of 1,964 ± 2 Ma (980–1122 °C fractions). Analyses (2) of the Encrucijada hornblende yield plateau ages of 1,972 ± 4 Ma (from 989–1025 °C fractions) and 1,969 ± 3 Ma (from 1000–1050 °C fractions).

The ages of the portions of the two biotite spectra corresponding to >60% release of $^{39}Ar_K$ are also quite similar to each other (see Fig. 4). The Harper biotite yields a plateau age of 1,894 ± 2 Ma (999–1250 °C fractions), and the Encrucijada biotite yields an integrated age of 1,883 ± 2 Ma (999–1185 °C fractions) for these segments of the spectra.

For both localities, the hornblende plateau ages are slightly younger than, although statistically indistinguishable from, the corresponding Rb-Sr whole-rock isochron ages of 2,064 ± 87 Ma and 1,983 ± 34 Ma, respectively. The Encrucijada Rb-Sr biotite age (1,958 ± 45 Ma) falls between its $^{40}Ar/^{39}Ar$ hornblende and biotite ages.

In the zero to 60% ^{39}Ar portions of the age spectra, both the Encrucijada hornblende and the biotite exhibit greater partial loss of ^{40}Ar compared to their Liberian counterparts, presumably because of later low-temperature thermal events. We defer discussion of this until Part II.

The main conclusions from the argon ages are: (1) the post–2.0-Ga *high-temperature* cooling histories of the Encrucijada pluton and the Harper amphibolite are indistinguishable, and (2) the argon data and the Rb-Sr ages are in agreement.

Downloaded from gsabulletin.gsapubs.org on March 6, 2010

TABLE 1. SOUTH AMERICAN AND AFRICAN PALEOMAGNETIC DATA

Shield formation	Abbr.*	Paleomagnetic data: No. of sites (samples)	Lat. (°N)	Long. (°E)	K	α_{95}	Radiometric ages	Refs.
Guayana	EnA1	5	55	188	150	6		1
Encrucijada		(20)	53	202	(60)	(4)		
	EnA2	3	-37	36	40	18		1
		(9)	-38	37	(15)	(15)		
							2,064 ± 87 Rb/Sr WR iso	2
							1,972 ± 4 $^{40}Ar/^{39}Ar$ hbde	1
							1,958 ± 45 Rb/Sr bio	3
							1,883 ± 2 $^{40}Ar/^{39}Ar$ bio	1
West African	HaA1	3	10	253	44	19		1
Harper		(11)	4	255	(39)	(7)		
	HaA2	(3)	3	92	(76)	(14)		1
							1,983 ± 34 Rb/Sr WR iso	4
							1,964 ± 2 $^{40}Ar/^{39}Ar$ hbde	1
							1,894 ± 2 $^{40}Ar/^{39}Ar$ bio	1
Kalahari								
Bushveld Complex	BC	5(103)†	23	36	40	12		5
							2,096 ± 12 $^{40}Ar/^{39}Ar$ WR†	6
							2,051 ± 24 Rb/Sr WR iso†	7
							2,050 ± 30 U/Pb zircon§	6
Vredefort Complex	VC	8(22)**	26	39	21	12		8
							2,006 ± 47 Rb/Sr min iso††	9
							1,928 Rb/Sr min iso§§	9
							1,950 ± 60 Rb/Sr WR iso	10
Kalahari								
Limpopo Belt Gneiss	LBG	12(85)***	21	27	(32)	(8)		11
							1,989 ± 40 Rb/Sr bio†††	12
							1,909 ± 50 Rb/Sr kspar	12

*Abbreviations used in Figures 8 and 9.
†Samples were collected from Bushveld gabbro.
§Samples were collected from Bushveld granophyre.
**Samples were collected from both the basic granophyre and the nepheline syenite.
††Samples were collected from the nepheline syenite.
§§Samples were collected from the basic granophyre.
***Group A directions.
†††Average of five Rb/Sr biotite ages reported from those localities yielding group A directions.
Note: references cited: 1. This study. 2. Posadas and Kalliokoski, 1967. 3. Short and Steenken, 1962. 4. Hurley and others, 1971. 5. Gough and Van Niekerk, 1959. 6. Coertze and others, 1978. 7. Hamilton, 1977. 8. Hargraves, 1970. 9. Nicolaysen and others, 1963. 10. Hart and others, 1981. 11. Morgan and Briden, 1981. 12. Van Breemen and Dodson, 1972.

TABLE 2. $^{40}Ar/^{39}Ar$ ANALYTICAL DATA

T(°C)	f_1	f_2	Atmos. (%)	^{40}Ar ($\times 10^{-6}$ cm^{3}STP g)	f(39)	$^{37}Ar/^{39}Ar$	Age	(Ma)*
				Sample Hbde ON33 (1st run)				
900	-0.001	0.001	44	12.32	0.157	0.80	1,706 ±	8
925	-0.002	0.015	16	3.76	0.042	3.55	1,855 ±	11
950	-0.003	0.039	8	7.43	0.078	4.48	1,926 ±	10
962	-0.003	0.006	34	0.44	0.005	5.04	1,746 ±	194
975	-0.003	0.030	10	1.28	0.013	4.67	1,945 ±	39
987	-0.003	0.034	10	1.88	0.019	4.79	1,950 ±	47
1000	-0.003	0.057	6	7.43	0.075	4.82	1,972 ±	8
1017	-0.003	0.063	5	39.07	0.397	4.80	1,966 ±	3
1025	-0.003	0.038	9	9.05	0.091	4.82	1,982 ±	9
1050	-0.003	0.012	21	0.87	0.009	4.87	1,967 ±	77
1075	-0.003	0.015	17	8.97	0.087	4.83	2,025 ±	8
1100	-0.004	0.001	42	1.50	0.011	6.64	2,396 ±	34
1125	-0.009	-0.001	42	1.66	0.009	14.53	2,858 ±	50
1150	-0.026	-0.010	50	1.36	0.006	40.66	2,976 ±	64
1150	-0.026	-0.021	71	0.47	0.002	40.21	3,176 ±	211
Mass = 0.0434								
Standard $(40/39)_s$† = 37.15 ± 0.03; age = 1,071 Ma								
				Sample Hbde ON33 (2nd run)				
800	-0.000	0.007	6	4.81	0.062	0.46	1,717 ±	19
825	-0.000	0.005	7	2.88	0.038	0.43	1,689 ±	25
850	-0.001	0.006	12	1.94	0.025	0.86	1,705 ±	30
875	-0.001	0.003	24	1.24	0.017	1.17	1,656 ±	87
900	-0.001	0.006	19	1.43	0.018	1.80	1,753 ±	54
925	-0.003	0.021	12	2.12	0.023	3.88	1,896 ±	32
950	-0.003	0.015	20	0.18	0.010	4.50	1,787 ±	48
985	-0.003	0.103	3	19.00	0.201	4.91	1,940 ±	7
989	-0.003	0.103	3	14.28	0.148	4.94	1,965 ±	6
996	-0.003	0.113	3	10.19	0.104	5.09	1,984 ±	8
1003	-0.003	0.066	5	8.85	0.091	4.98	1,978 ±	15
1009	-0.003	0.045	8	6.65	0.067	5.07	1,971 ±	14
1015	-0.003	0.015	18	0.74	0.003	4.57	1,932 ±	160
1025	-0.003	0.041	9	8.89	0.092	5.20	1,970 ±	8
1075	-0.003	0.031	10	7.11	0.071	4.99	2,010 ±	11
1150	-0.008	0.018	20	3.77	0.025	12.37	2,537 ±	26
Mass = 0.0373								
Standard = $(40/39)_s$† = 37.15 ± 0.03; age = 1.071 Ma								

Downloaded from gsabulletin.gsapubs.org on March 6, 2010

TABLE 2. (*Continued*)

T(°C)	f_1	F_2	Atmos. (%)	^{40}Ar ($\times 10^{-6}$ cm^{3}STP g)	f(39)	$^{37}Ar/^{39}Ar$	Age	(Ma)*
				Sample Bio ON33				
508	0.0000	-0.0000	39	1.28	0.003	0.00	1,358 ±	53
550	-0.0001	0.0003	33	0.85	0.002	0.20	1,369 ±	83
575	-0.0001	0.0002	31	0.73	0.002	0.09	1,477 ±	83
600	-0.0001	0.0004	19	1.34	0.003	0.10	1,458 ±	79
625	-0.0000	0.0004	10	2.26	0.004	0.06	1,603 ±	68
650	-0.0000	0.0001	12	3.92	0.007	0.02	1,595 ±	16
675	-0.0000	0.0006	6	7.50	0.013	0.05	1,680 ±	18
700	-0.0000	0.0002	4	12.24	0.020	0.01	1,737 ±	6
733	-0.0000	0.0002	2	15.37	0.024	0.01	1,789 ±	4
766	-0.0000	0.0006	1	33.01	0.049	0.01	1,814 ±	2
800	-0.0000	0.0005	1	49.43	0.074	0.01	1,824 ±	1
850	-0.0000	0.0013	1	78.97	0.116	0.01	1,837 ±	1
900	-0.0000	0.0030	1	100.24	0.145	0.03	1,854 ±	1
950	-0.0001	0.0099	1	87.10	0.125	0.11	1,861 ±	2
1000	-0.0001	0.0118	1	88.87	0.126	0.18	1,874 ±	2
1050	-0.0001	0.0039	2	80.41	0.114	0.10	1,879 ±	1
1051	-0.0000	0.0001	10	21.39	0.030	0.02	1,873 ±	5
1083	-0.0000	0.0001	4	43.68	0.061	0.01	1,892 ±	3
1116	-0.0000	0.0000	6	33.05	0.046	0.00	1,888 ±	3
1150	-0.0000	0.0000	11	18.75	0.026	0.00	1,911 ±	4
1183	-0.0000	-0.0000	36	7.05	0.010	0.05	1,916 ±	12
Mass = 0.0166 Standard $(40/39)_s$† = 46.81 ± 0.07; age = 1,071 Ma								
				Sample Bio ON239				
850	-0.002	0.002	38	2.02	0.026	2.64	1,947 ±	7
900	-0.002	0.002	38	0.89	0.011	2.92	2,010 ±	40
925	-0.002	0.005	30	0.74	0.009	3.48	1,958 ±	37
950	-0.003	0.012	21	1.14	0.015	4.73	1,909 ±	25
980	-0.004	0.066	7	3.94	0.050	6.17	1,958 ±	9
1000	-0.004	0.109	4	6.86	0.086	6.52	1,973 ±	6
1012	-0.004	0.113	4	8.61	0.108	6.60	1,965 ±	3
1025	-0.004	0.115	4	9.99	0.126	6.65	1,961 ±	3
1037	-0.004	0.110	4	10.24	0.129	6.67	1,964 ±	4
1050	-0.004	0.071	6	7.26	0.092	6.64	1,956 ±	5
1062	-0.004	0.063	7	7.44	0.093	6.64	1,966 ±	3
1075	-0.004	0.050	9	5.92	0.074	6.59	1,969 ±	5
1100	-0.004	0.038	11	5.81	0.073	6.61	1,971 ±	6
1120	-0.004	0.020	19	3.88	0.049	6.86	1,959 ±	7
1150	-0.005	0.015	24	2.93	0.035	7.96	2,011 ±	6
1175	-0.005	0.006	36	2.05	0.025	8.04	1,996 ±	12
Mass = 0.0458 Standard $(40/39)_s$† = 36.77 ± 0.05; age = 1,071 Ma								
				Sample Bio 239				
500	-0.0000	0.0005	32	3.29	0.013	0.090	793 ±	8
550	-0.0002	0.0042	12	3.04	0.008	0.311	1,102 ±	3
575	-0.0002	0.0164	2	4.74	0.006	0.287	1,760 ±	14
600	-0.0000	0.0014	3	0.50	0.012	0.056	1,862 ±	6
625	-0.0000	0.0008	2	17.59	0.022	0.015	1,869 ±	2
650	-0.0000	0.0005	1	32.72	0.040	0.007	1,869 ±	1
675	-0.0000	0.0005	1	40.14	0.049	0.004	1,872 ±	2
700	-0.0000	0.0007	1	51.91	0.063	0.004	1,875 ±	1
733	-0.0000	0.0006	1	67.13	0.081	0.004	1,881 ±	1
766	-0.0000	0.0000	22	71.75	0.085	0.005	1,909 ±	2
800	-0.0000	0.0003	3	47.07	0.056	0.012	1,898 ±	2
900	-0.0000	0.0030	1	44.19	0.052	0.036	1,912 ±	1
950	-0.0000	0.0066	1	75.73	0.090	0.045	1,907 ±	1
1000	-0.0001	0.0136	1	102.15	0.122	0.091	1,895 ±	1
1050	-0.0001	0.0246	1	87.69	0.105	0.162	1,893 ±	1
1225	-0.0001	0.0038	4	161.61	0.193	0.209	1,895 ±	1
1250	-0.0002	0.0016	13	1.44	0.001	0.295	1,903 ±	33
1300	-0.0003	-0.0000	55	0.15	0.0001	0.422	2,058 ±	353
1600	-0.0000	-0.0000	81	0.31	0.0004	0.044	1,850 ±	156
Mass = 0.0301 Standard $(40/39)_s$† = 36.77 ± 0.05; age = 1,071 Ma								

Note: $f_1 = 1/[1-(37/39)_{Ca}/(37/39)_M]$, and $f_2 = f_1[1-(36/39)_{Ca}/36/39)_M]$, where $()_{Ca}$ = in argon isotope ratio gas extracted from irradiated calcium salts and $()_M$ = argon isotope ratio in gas extracted from irradiated mineral. f(39) = fraction of ^{39}Ar released in each step.

*$\lambda_\varepsilon = 0.581 \times\times 10^{-10}$ yr^{-1}; $\lambda_\beta = 4{,}961 \times 10^{-10}$ yr^{-1}; $^{40}K/K = 0.0116$ atom % (Steiger and Jäger, 1977).

†Weighted mean of three fusions.

Paleomagnetism

Of the 38 Encrucijada samples, 37 yielded stable magnetizations (as defined above) between 20 and 70 mT. These 37 samples constitute 2 distinct groups:

Group A. A distinct northwest-southeast, bipolar cluster was formed by 32 samples. Of these, 29 met the minimum quality criteria established above, and they are illustrated in Figure 2.

Group B. Directions distinct from group A were yielded by 5 samples, but they were similar to the secondary components removed from group A samples by as much as 20 mT AF demagnetization.

Of the Harper samples, 16 yielded stable magnetizations between 20 and 70 mT. These also constitute two distinct groups:

Group A. An east-west, bipolar cluster was formed by 14 samples. All of these samples met

Downloaded from gsabulletin.gsapubs.org on March 6, 2010

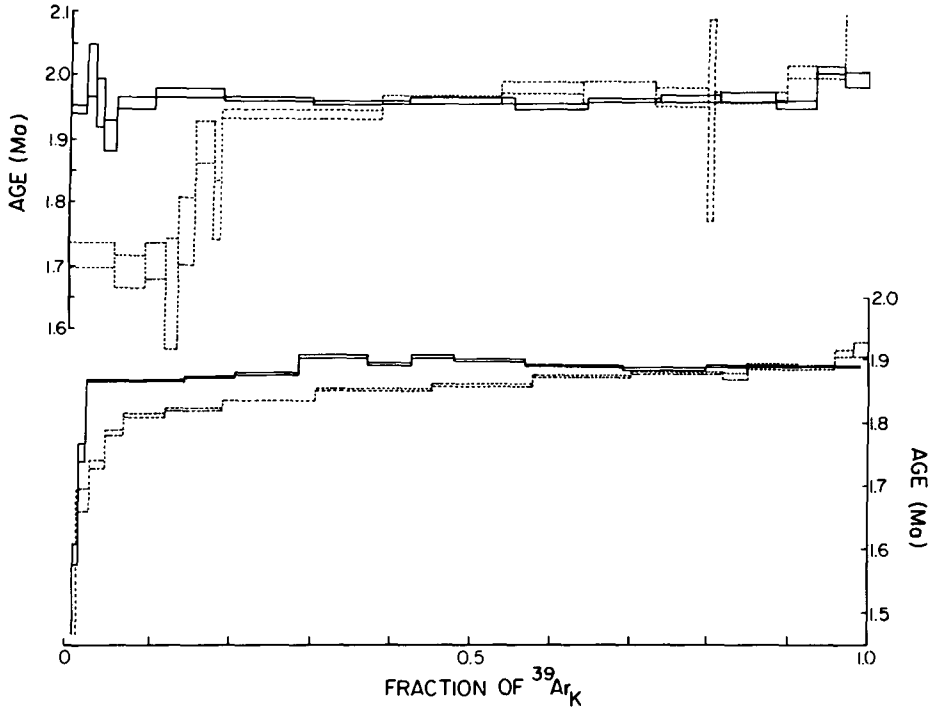


Figure 4. $^{40}Ar/^{39}Ar$ age spectra for the hornblendes (upper plot) and biotites (lower plot) of the Encrucijada pluton (dashed lines) and the Harper amphibolites (solid lines).

the minimum quality criteria, and they are illustrated in Figure 7 (see below).

Group B. Three samples are distinct from group A but are similiar to the secondary components removed from group A samples by as much as 20 mT AF demagnetization.

A bipolar remanence, therefore, is found for both the Encrucijada and Harper samples, which is stable with respect to both AF and thermal demagnetization (see Figs. 5 and 6). In the Encrucijada pluton, this remanence consists of a tightly clustered, northwesterly, horizontal set of vectors (Dec = 324, Inc = 7, k = 64, a_{95} = 4, n = 20), which we refer to as En_{A1}, and a less concentrated, east-southeasterly, negative inclination direction (Dec = 128, Inc = -23, k = 15,

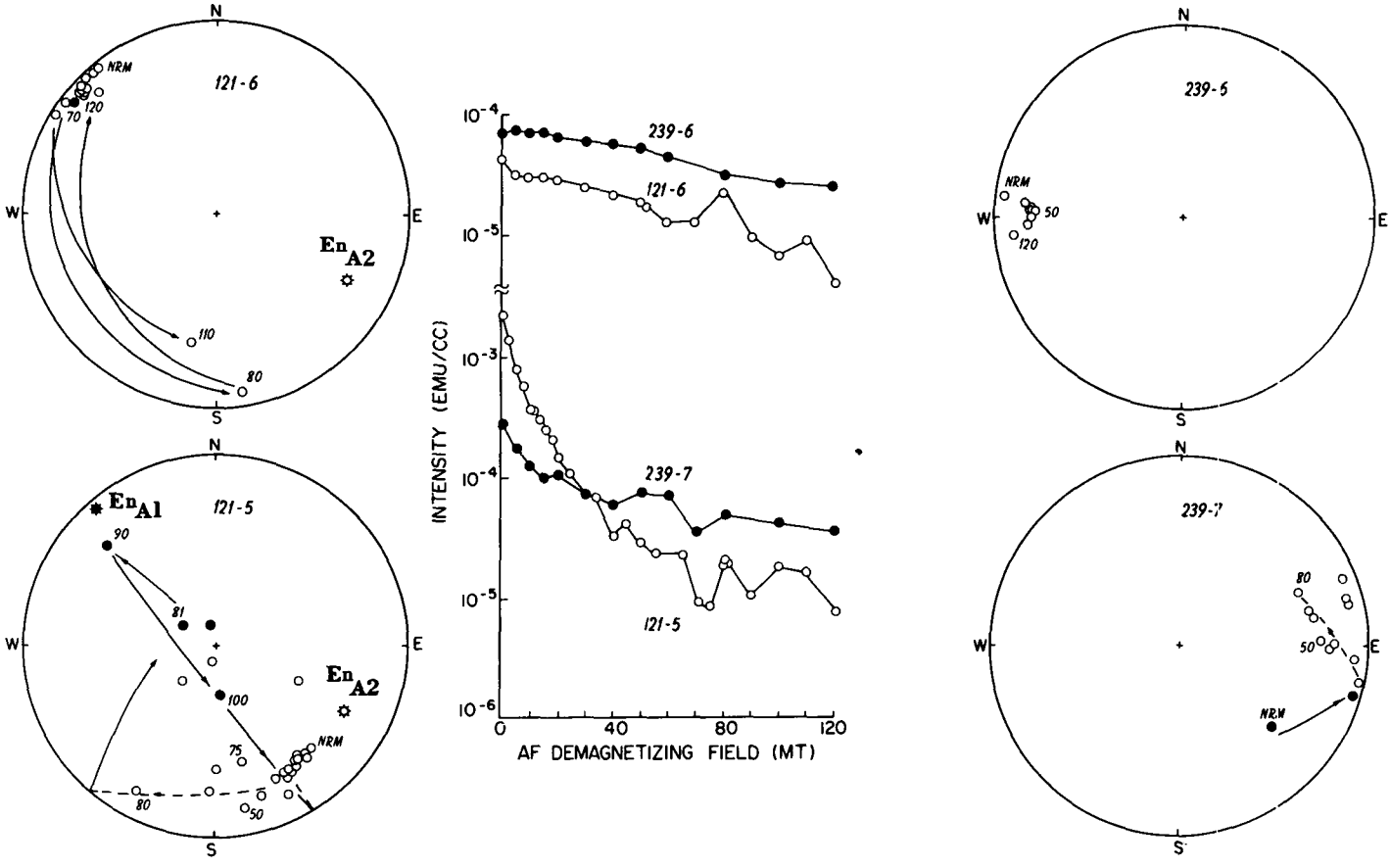


Figure 5. Representative thermal demagnetization data for two samples (121-5 and 6) from the Encrucijada pluton and two samples (239-6 and 7) from the Harper amphibolite. The En_{A1} remanence dominates the magnetization of sample 121-6, but En_{A2} may be present at temperature steps greater than 570 °C. The En_{A2} remanence dominates the magnetization of sample 121-5, but En_{A1} may be present at temperature steps greater than 590 °C. Similar behavior is exhibited by the Harper samples. The unblocking temperatures for all samples overlap and lie between 560 and 600 °C.

Downloaded from gsabulletin.gsapubs.org on March 6, 2010

a_{95} = 15, n = 9), which we refer to as En_{A2} (see Fig. 7). In the Harper amphibolite, this remanence consists of very tightly grouped, westerly, negative inclination directions (Dec = 274, Inc = -14, k = 39, a_{95} = 7, n = 11), which we refer to as Ha_{A1}, and tightly grouped, east-northeasterly, negative inclination directions (Dec = 89, Inc = -19, k = 76, a_{95} = 14, n = 3), which we refer to as Ha_{A2} (see Fig. 7). In the Encrucijada pluton and the Harper amphibolite, samples with predominantly Al vectors occur within 2 m of samples with A2 vectors within a single outcrop.

Analyses of thin sections (see Appendix 1) and magnetic studies of high-purity mineral separates and whole-rock samples (details presented in Part II) indicate the following.

1. The A components of the Encrucijada pluton are carried by magnetite grains that exhibit an intensively oxidized exsolution texture characteristic of high-temperature (>400 °C) pseudomorphic oxidation.

2. The A components of the Harper amphibolite are carried by rodlike magnetic inclusions in hornblende, which presumably formed at high temperature.

Thin-section and thermomagnetic analyses indicate that very minor low-temperature alteration is present in samples containing A components. From these observations, we feel it is reasonable to assume that the A components were acquired upon cooling through their blocking temperatures.

As seen in Figure 5, the unblocking temperatures of En_{A1} and Ha_{A1} (560–580 °C) coincide with those of En_{A2} and Ha_{A2} (560–600 °C) blocking temperatures. The thermal demagnetization data combined with the $^{40}Ar/^{39}Ar$ and Rb-Sr age data suggest that the ages of En_{A1}, En_{A2}, Ha_{A1}, and Ha_{A2} are indistinguishable.

The origin of bipolarity of the high-temperature remanence is difficult to explain, because both polarities frequently occur within the same outcrop yet have essentially the same blocking temperatures. The demagnetization data at higher temperatures (>580 °C in Fig. 5) and higher AF fields (>70 mT in Fig. 6) suggest that both polarities may, in fact, be present in the same specimen, but that one predominates. This latter conclusion is tentative, however, because the quality of the data deteriorates at higher demagnetization levels. Possible mechanisms that could cause the observed bipolar remanences are: (1) the magnetizations record reversals in the Earth's magnetic field; (2) one of the magnetizations is self-reversed; (3) one of the magnetizations is a secondary chemical remanent magnetization. These are discussed in detail in our second paper.

Regardless of the mode of origin, at least one of the high-temperature remanences probably was acquired as these two rock units cooled through their magnetite blocking temperatures after peak metamorphism. In that the cooling rate was ~15 °C/Ma and the blocking temperature interval at least 20 °C (see Fig. 5), the magnetizations probably have averaged ~10^5 to 10^6 yr of secular variation and therefore can be presumed to provide reliable paleomagnetic (paleogeographic) poles.

DISCUSSION

Venezuelan and Liberian Pole Dates

When South America is restored in a Bullard and others (1965) reconstruction (see Fig. 8), the Encrucijada A1 and A2 poles and the Harper A1 and A2 poles lie in proximity. The

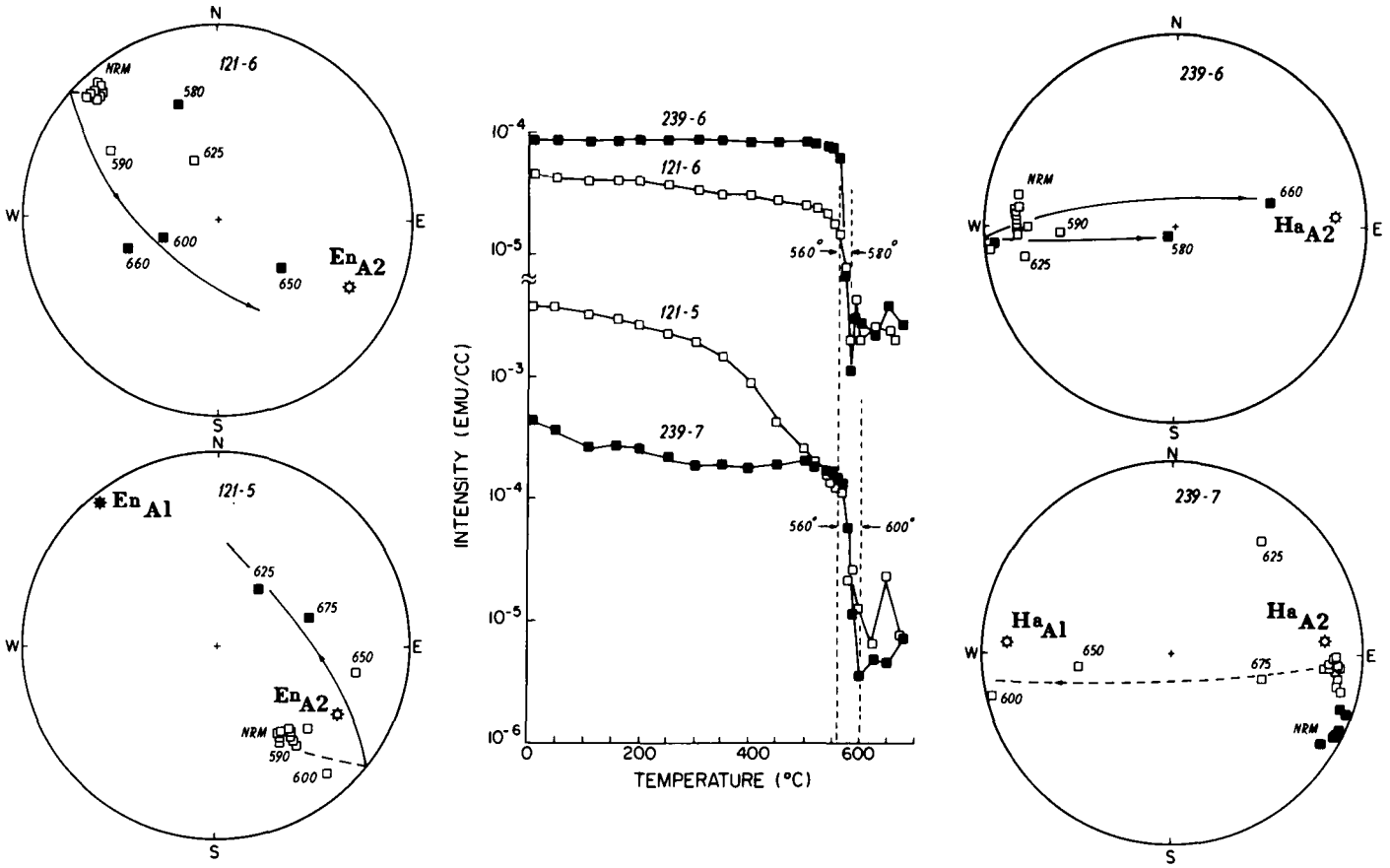


Figure 6. Representative AF demagnetization data for specimens from the same samples as in Figure 5. The En_{A1} component dominates the magnetization of sample 121-6 to 70 mT but at higher fields shows a tendency to move toward an En_{A2} direction. Similarly, in sample 121-5, the En_{A2} component dominates the magnetization to 75 mT but at higher fields shows a tendency to move toward an En_{A1} direction.

Downloaded from gsabulletin.gsapubs.org on March 6, 2010

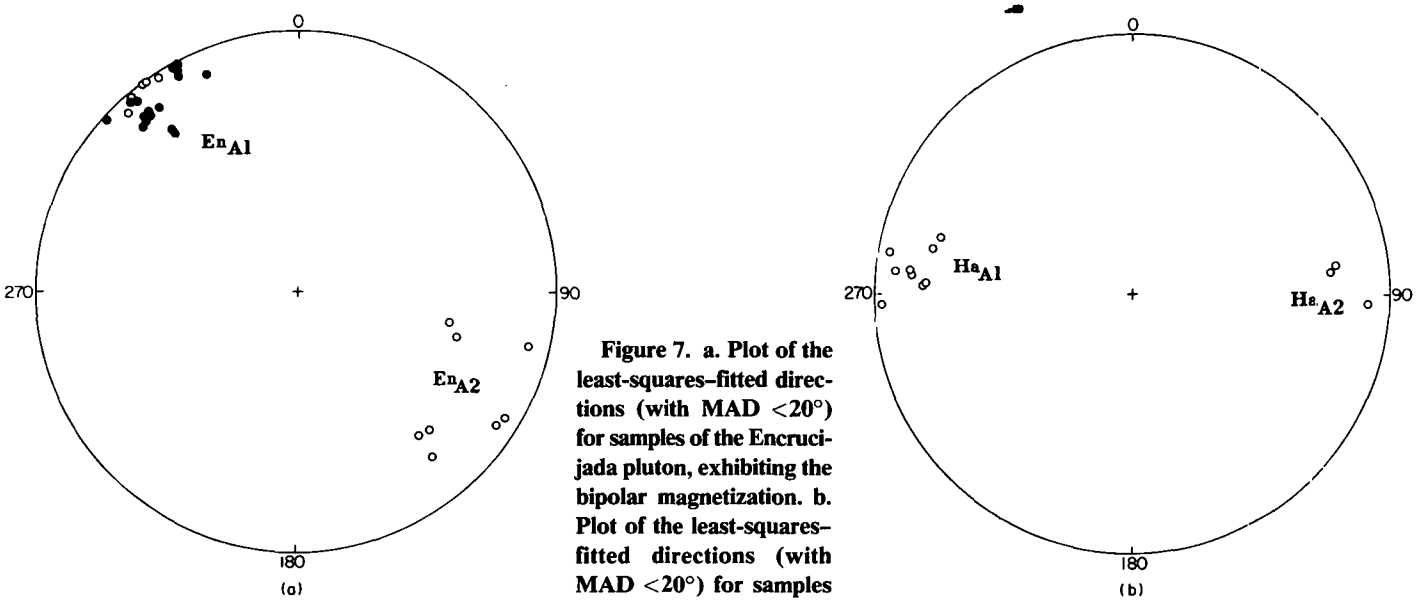


Figure 7. a. Plot of the least-squares–fitted directions (with MAD <20°) for samples of the Encrucijada pluton, exhibiting the bipolar magnetization. b. Plot of the least-squares–fitted directions (with MAD <20°) for samples of the Harper amphibolites, exhibiting a bipolar magnetization very similar to that of the Encrucijada samples.

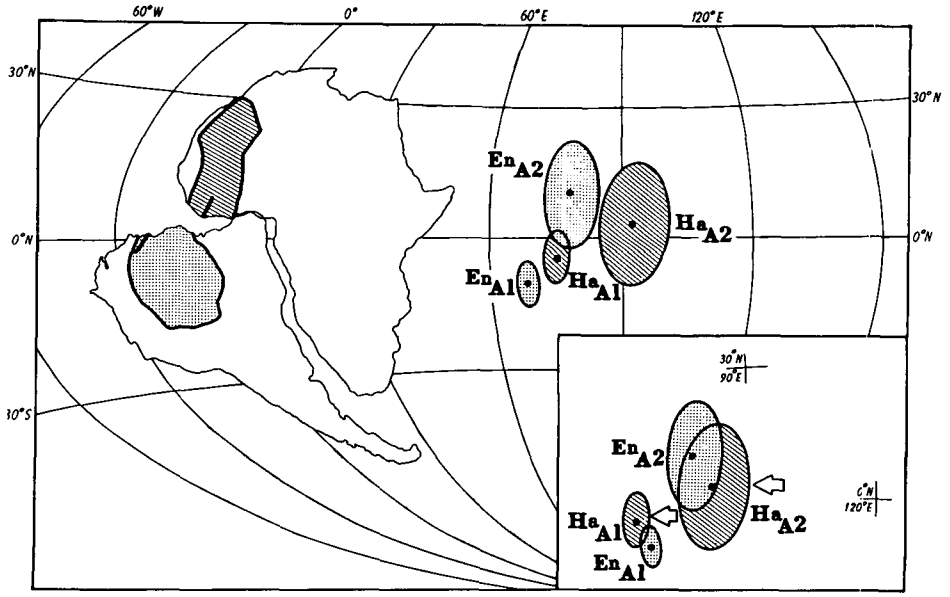


Figure 8. South America and Africa in the Bullard and others (1965) reconstruction. The Encrucijada paleomagnetic poles, En_{A1} and En_{A2}, and their 95% confidence ovals (shaded) have been rotated using the Bullard and others (1965) rotation pole. The Harper paleomagnetic poles, Ha_{A1} and Ha_{A2}, and their 95% confidence ovals (hachured) are shown. The inset shows the relative position of the two sets of poles after the West African Shield has been moved westward with respect to the Guayana Shield to align the Guri and Dube fault zones.

Harper A1 and A2 poles are displaced, however, by ~10° to the east of the Encrucijada A1 and A2 poles. A similar offset was noted by Onstott and Hargraves (1981) for the paleomagnetic poles from 1.7-Ga diabases in Venezuela and Guayana, when compared to paleomagnetic poles from diabases of similar age in Ivory Coast and Algeria. By shifting the West African Shield with respect to the Guayana Shield by movement along the Pan-African Belt so as to align the Guri and Dube fault zones (GFZ and DFZ in Fig. 1), these diabase poles could be brought into agreement. This same shift also brings the Encrucijada and Harper poles into better agreement (see Fig. 8).

The location of the Harper amphibolite, on the opposite side of the Pan-African Belt from the Encrucijada pluton and only 50 km east of the Pan-African Belt itself, supports the hypothesis of Onstott and Hargraves (1981) that the zone of proposed dislocation occurs within the Pan-African mobile belt. Additional well-dated paleomagnetic poles, however, showing the same offset will be required before that hypothesis can be considered more than tentative.

Buchan and others (1977) and Berger and York (1979a, 1979b, 1981a) presented evidence to show that in plutonic rocks, the argon blocking temperatures of hornblende and biotite typically straddle the magnetic blocking temperature of magnetite. This suggests that the ages of the A1 and/or A2 magnetizations for both the Encrucijada pluton and the Harper amphibolite lie between 1.89 and 1.96 Ga. If, in fact, the A1 and A2 magnetizations were acquired before the hornblende argon clocks were switched on, then the ages of these magnetizations lie between the 1.96- to 1.97-Ga hornblende ages and the 1.98- and 2.06-Ga Rb-Sr whole-rock isochron ages. Without detailed argon geothermometry, we can confidently conclude, therefore, that the ages of the high-temperature remanences from both Venezuela and Liberia must lie between 1.89 and 2.06 Ga and are identical within this range of uncertainty. A more precise argon thermochronometric calibration of the age of magnetization is presented in Part II.

Venezuelan, Liberian, and South African Poles

If we now compare the Venezuelan and Liberian poles with South African poles of approximately equivalent age (see Table 1), we find that the South African (Kalahari Shield) results

Downloaded from gsabulletin.gsapubs.org on March 6, 2010

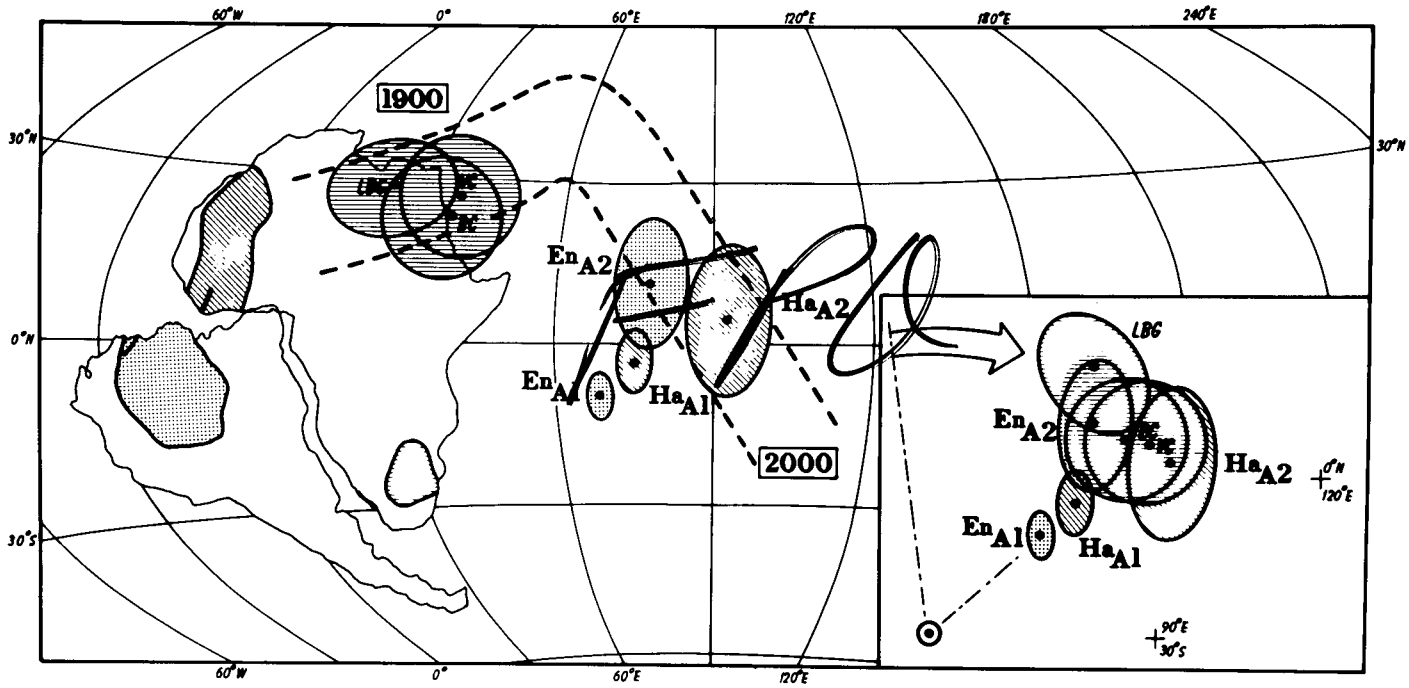


Figure 9. South America and Africa in the Bullard and others (1965) reconstruction. Shown are the Encrucijada and Harper poles (positions and symbols are the same as for Fig. 8), the Kalahari Shield poles and 95% confidence ovals (horizontal hachure), the Bushveld Complex (BC), the Vredefort Dome (VC), and the Limpopo Belt Gneiss (LBG). The dashed apparent polar wander swathe and attached ages (in millions of years) are from Piper (1982). The inset illustrates the excellent juxtaposition of the Kalahari poles on the West African and Guayana poles obtained when the Kalahari Shield is rotated according to the reconstruction of Onstott and Hargraves (1981) (rotation pole 44°E, 27°S, and 64°CW) to align the Limpopo and Ubendian mobile belts. The polarity of poles BC and VC matches the polarity of poles En_{A2} and Ha_{A2}.

are distinct (BC, VC, and LBG in Fig. 9). To resolve this discrepancy, one must postulate either of the following.

1. The two groups of poles are different enough in age to be compatible with a single apparent polar wander path for all of Africa and South America.

2. The two groups of poles are identical in age, and thus the relative positions of the African Shields were considerably different during this period of the Precambrian.

Piper (1982) espoused the former viewpoint by claiming that these Kalahari Shield paleomagnetic poles (BC, VC, and LBG) are 1.9 Ga in age and, therefore, younger than the Guayana Shield poles of Onstott and Hargraves (1981) and that both sets of poles are consistent with his Proterozoic Pangea polar wander curve (see Fig. 9).

In contrast, Onstott and Hargraves (1981) claimed that the 1,800- to 2,200-Ma paleomagnetic data from the Guayana, West African, and Congo Shields and from the Richtersveld Province formed a self-consistent data set that is distinct from the Kalahari Shield paleomagnetic poles; rotation of the Kalahari Shield relative to these shields is required to remove the discrepancy.

The difference in the interpretation is caused by the difference in the age assigned to the Kalahari Shield paleomagnetic poles by Piper (1982), on the one hand, and Onstott and Hargraves (1981), on the other. The pertinent radiometric-age information available for the Kalahari Shield paleomagnetic poles is summarized below.

Bushveld Complex. Rb-Sr whole-rock (Hamilton, 1977) and $^{40}Ar/^{39}Ar$ whole-rock (Coertze and others, 1978) analyses of the Bushveld gabbro yield 2,051 ± 24 and 2,096 ± 12 Ma, respectively. U-Pb zircon analyses yield an age of 2,050 ± 30 Ma for the Bushveld granophyre, which overlies the gabbro. The complex is intruded by the Nebo (Red) granite (Walraven, 1977), which yields a Rb-Sr whole-rock isochron age of 1,913 ± 30 Ma (Davies and others, 1970) and a U-Pb zircon age of 1,920 ± 40 Ma (Coertze and others, 1978). Piper (1982) quoted a 1910 Ma Rb-Sr date for the Bushveld Complex paleomagnetic pole, which we presume he derived from the Rb-Sr isochron age of the Nebo granite. The paleomagnetic samples, however, were collected from the Main Zone of the Bushveld gabbro, not from the granite (Gough and Van Niekerk, 1959). In that the $^{40}Ar/^{39}Ar$ age of 2,096 Ma suggests this portion of the gabbro has not been significantly reheated by the Nebo granite, an age of 2,050 to 2,100 Ma would seem more appropriate for the Bushveld pole.

Vredefort Dome. The basic granophyre and the nepheline syenite of the Vredefort Dome yield Rb-Sr mineral isochron ages of 1,938 Ma and 2,006 ± 47 Ma, respectively (Nicolaysen and others, 1963). These ages are consistent with the Rb-Sr whole-rock isochron age of 1,950 ± 60 Ma reported for the Central Intrusive Granite (Hart and others, 1981). The Vredefort paleomagnetic pole is based upon samples collected from the basic granophyre ring dikes (Hargraves, 1970). The slopes of the mineral isochrons are strongly weighted by the Rb-Sr analyses of biotites, which have an isotopic closure temperature less than or equal to the magnetic blocking temperature of magnetite, depending on the cooling rate. The Rb-Sr whole-rock and mineral results, therefore, suggest an approximate age range of 1.9 to 2.0 Ga for the Vredefort Complex paleomagnetic pole.

Limpopo Belt Gneiss. Paleomagnetic studies by Morgan and Briden (1981) yielded a pole (pole A) quite similar to the Bushveld and Vredefort poles (see Fig. 9). Rb-Sr biotite analyses of these rocks (Van Breemen and Dodson, 1972) yield an average of 1,989 ± 40 Ma. If the A magnetization was acquired upon uplift and cooling of the Limpopo Belt gneisses, then the Rb-Sr results provide a minimum age for the magnetization, which is significantly older than

Downloaded from gsabulletin.gsapubs.org on March 6, 2010

the 1.9-Ga age inferred from Piper's (1982) polar wander path (see Fig. 9).

In summary, the 1.94- to 2.05-Ga range of ages reported for the Bushveld Complex, the Vredefort Complex, and the Limpopo Belt gneisses of the Kalahari Shield overlaps the 1.89- to 2.0-Ga age range for the Guayana and West African poles. Furthermore, the radiometric age data do not appear to be consistent with a 1.9-Ga age as given by Piper (1982). Although this approximate equivalence of the age of the Encrucijada and Harper A1 and A2 poles to these South African poles lends support to the rotational hypothesis of Onstott and Hargraves (1981), their hypothesis cannot be rigorously tested until the precise ages of the magnetizations of the Kalahari units have been determined.

CONCLUSIONS

In this study, we have obtained paleomagnetic poles that are virtually identical in age from two rock units on opposite sides of the Liberian Pan-African Belt. These poles are offset ~10° in a manner remarkably consistent with the structural offset between the Guri fault zone of the Guayana Shield and the Dube fault zone of the West African Shield. Onstott and Hargraves (1981) proposed that the offset in the Precambrian paleomagnetic data between the Guayana and West African Shields could have been caused by a northwestward transcurrent motion along the Brazilian mobile belt of the Guayana Shield with respect to the West African and Congo Shields. An equally plausible alternative, however, would be an eastward transcurrent motion of ~1,000 km along the Pan-African Belt of the West African Shield with respect to the Guayana Shield. Such a motion could be related to the crustal shortening observed in the Pharusian Pan-African Belt (Caby and others, 1981). Until precisely determined and well-dated paleomagnetic poles from the Congo Shield are available, however, it is not possible to determine which hypothesis is more likely.

If, as we believe, the paleomagnetic poles of the Kalahari Shield are the same age as the Guayana and West African poles reported here, then the Kalahari Shield must have been displaced with respect to the Guayana and West African Shields. The Kalahari rotation hypothesis of Onstott and Hargraves (1981), which involves transcurrent motions along the Namaqua front, was dictated by use of some uncertain paleomagnetic data from Namaqualand and is not necessarily correct. It represents only one possible model that would resolve this discrepancy between the two sets of poles.

ACKNOWLEDGMENTS

We acknowledge the assistance of Naoma Dorety and Bruce Moskowitz of Princeton University in the collection of the paleomagnetic and rock magnetic data. Peter Kuybida and John Kenyon (University of Toronto) assisted in the collection and analysis of the argon data. L. S. Hollister, Charles Kulick, and Will Maze (Princeton University) provided the microprobe analyses. D. Kent (Lamont-Doherty Geological Observatory) and D. J. Dunlop (University of Toronto) allowed use of their thermal demagnetization equipment. Dr. A. Bellizia, N. Benaim, and V. Tepidino of the Ministerio de Energia of Venezuela provided financial and logistical support for the field work in the Guayana Shield. Jenkins Dorbor and Dr. Eugene Shannon, of the Liberian Geological Survey, gave guidance in sample collection, support in the field, and hospitality during the brief stay in their country. The funding for this research was provided by National Science Foundation Grant no. EAR 80-08207 and by the National Science and Engineering Research Council of Canada. We thank W. A. Morris and an anonymous reviewer for their constructive comments. Last, but not least, we acknowledge A. Palmer for her perseverance during the prolonged absences of the senior author in the course of this study.

REFERENCES CITED

Berger, G. W., and York, D., 1979a, Calibration of Grenvillian paleopoles by $^{40}Ar/^{39}Ar$ dating: Nature, v. 277, p. 46–47.

——1979b, $^{40}Ar/^{39}Ar$ dating of multicomponent magnetizations in the Archean Shelley Lake granite, northwestern Ontario: Canadian Journal of Earth Sciences, v. 16, p. 1933–1941.

——1981a, $^{40}Ar/^{39}Ar$ dating of the Thanet gabbro, Ontario: Looking through the Grenvillian metamorphic veil and implications for paleomagnetism: Canadian Journal of Earth Sciences, v. 18, p. 266–275.

——1981b, Geothermometry from $^{40}Ar/^{39}Ar$ dating experiments: Geochimica et Cosmochimica Acta, v. 45, p. 795–811.

Brock, M. R., Chidester, A. H., and Baker, M.W.G., 1977, Geologic map of the Harper Quandrangle, Liberia: Map I 780-D (prepared by U.S. Geological Survey, Reston, Virginia).

Buchan, K. L., Berger, G. W., McWilliams, M. O., York, D., and Dunlop, D. J., 1977, Thermal overprinting of natural remanent magnetization and K/Ar ages in metamorphic rocks: Journal of Geomagnetism and Geoelectricity, v. 29, p. 401–410.

Bullard, E. C., Everett, J. E., and Smith, A. G., 1965, Fit of continents around the Atlantic, *in* Blackett, E. C., Bullard, E. C., and Runcorn, S. K., eds., A symposium on continental drift: Royal Society of London Philosophical Transactions, ser. A., v. 258, p. 41–75.

Caby, R., Bertrand, J.M.L., and Black, R., 1981, Pan-African ocean closure and continental collision in the Hoggar-Iforas segment, central Sahara, *in* Kroner, A., ed., Precambrian plate tectonics: Amsterdam, Elsevier Scientific Publishing Company, p. 407–434.

Cahen, L., 1961, Review of geochronological knowledge in middle and northern Africa: New York Academy of Sciences Annals, v. 91, p. 535–567.

Coertze, F. J., Burger, A. J., Walraven, F., Marlow, A. G., and McCaskie, D. R., 1978, Field relations and age determinations in the Bushveld Complex: Geological Society of South Africa Transactions, v. 81, p. 1–11.

Creer, K. M., and Sanver, M., 1963, The use of the sun compass, *in* Collinson, D. W., Creer, K. M., and Runcorn, S. K., eds., Methods in paleomagnetism: Amsterdam, Elsevier Publishing Company, p. 11–20.

Davies, R. D., Allsopp, H. L., Erlank, A. J., and Manton, W. T., 1970, Sr-isotopic studies on various layered mafic intrusions in southern Africa, *in* Symposium on the Bushveld Igneous Complex and other layered intrusions: Geological Society of South Africa Special Publication, no. 1, p. 576–593.

Doell, R. R., and Cox, A., 1963a, Paleomagnetic sampling with a portable coring drill, *in* Collinson, D. W., Creer, K. M., and Runcorn, S. K., eds., Methods in paleomagnetism: Amsterdam, Elsevier Publishing Company, p. 21–25.

——1963b, Recording magnetic balance, *in* Collinson, D. W., Creer, K. M., and Runcorn, S. K., eds., Methods in paleomagnetism: Amsterdam, Elsevier Publishing Company, p. 440–441.

Elmore, W. C., 1938, Ferromagnetic colloid for studying magnetic structures: Physics Review, v. 54, p. 309–310.

Farrar, E., Mcintyre, R. M., York, D., and Kenyon, W. J., 1964, A simple mass spectrometer for the analysis of argon at ultra-high vacuum: Nature, v. 204, p. 531.

Fisher, R. A., 1953, Dispersion on a sphere: Royal Society of London, Proceedings, v. A217, p. 295–305.

Gibbs, A. K., 1975, Geology of the Barama-Mazaruni supergroup of Guyana [Ph.D. dissert.]: Cambridge, Massachusetts, Harvard University, 387 p.

Gough, D. I., and Van Niekerk, C. B., 1959, A study of the paleomagnetism of the Bushveld Gabbro: Philosophy Magazine, v. 4, p. 196–213.

Hall, C. M., 1982, The application of K-Ar and $^{40}Ar/^{39}Ar$ methods to the dating of Recent volcanics and the Laschamp Event [Ph.D. thesis]: Toronto, Ontario, University of Toronto, 186 p.

Hamilton, J., 1977, Sr isotopic and trace element studies of the Great Dyke and Bushveld mafic phases and their relation to early Proterozoic magma genesis in southern Africa: Journal of Petrology, v. 18, p. 24–52.

Hanes, J. A., 1979, An $^{40}Ar/^{39}Ar$ geochronological study of Precambrian diabase dikes [Ph.D. thesis]: Toronto, Ontario, University of Toronto, 240 p.

Hargraves, R. B., 1970, Paleomagnetic evidence relevant to the origin of the Vredefort Ring: Journal of Geology, v. 78, p. 253–263.

Hart, R. J., Welke, H. J., and Nicolaysen, L. O., 1981, Geochronology of the deep profile through Archean basement at Vredefort, with implication for early crustal evolution: Journal of Geophysical Research, v. 86, p. 10663–10680.

Hedge, C. E., Marvin, R. F., and Naeser, C. W., 1975, Age provinces in the basement rocks of Liberia: U.S. Geological Survey Journal of Research, v. 3, p. 1229–1242.

Hillhouse, J. W., 1977, A method for the removal of rotational remanent magnetization acquired during alternating field demagnetization: Royal Astronomical Society Geophysical Journal, v. 50, p. 29–34.

Hurley, P. M., 1973, On the origin of the 450 ± 200 m.y. orogenic belts, *in* Tarling, D. H., and Runcorn, S. K., eds., Implications of continental drift to the Earth sciences, 2: London, Academic Press, p. 1083–1089.

Hurley, P. M., and Rand, J. R., 1969, Pre-drift continental nuclei: Science, v. 164, p. 1229–1242.

Hurley, P. M., Leo, G. W., White, R. W., and Fairbairn, H. W., 1971, The Liberian age province (ca. 2700 m.y.) and adjacent provinces in Liberia and Sierra Leone: Geological Society of America Bulletin, v. 82, p. 3483–3490.

Kalliokoski, J., 1965, Geology of north-central Guayana Shield, Venezuela: Geological Society of America Bulletin, v. 76, p. 1027–1049.

Kirschvink, J. L., 1980, The least-squares line and plane and the analysis of paleomagnetic data: Royal Astronomical Society Geophysical Journal, v. 62, p. 699–718.

Kolbe, P., Pinson, W. H., Jr., Saul, J. M., and Miller, E. W., 1967, Rb-Sr study on country rocks of the Bosumtwi Geochimica et Cosmochimica Acta, v. 31, p. 869–875.

Lopez-Martinez, M., 1982, $^{40}Ar/^{39}Ar$ studies and the cooling history of the Grenville Province [M.S. thesis]: Toronto, Ontario, University of Toronto, 107 p.

Masliwec, A., 1981, The direct dating of ore minerals [M.S. thesis]: Toronto, Ontario, University of Toronto, 119 p.

Montgomery, C. W., Hurley, P. M., Fairbairn, H. W., and Gaudette, H. E., 1977, Equilibrated domains and combined Rb-Sr and U-Pb systematics in the history of a granulite: Massachusetts Institute of Technology Geochronology Laboratory Twenty-first Progress Report for 1974–1976, p. 1–25.

Morgan, G. E., and Briden, J. C., 1981, Aspects of Precambrian paleomagnetism with new data from the Limpopo belt and Kaapvaal cratons in southern Africa: Physics of the Earth and Planetary Interiors, v. 24, p. 142–168.

Nicolaysen, L. O., Burger, A. J., and Van Niekerk, C. B., 1963, The origin of the Vredefort Dome structure in the light of new isotopic data [abs.]: International Union of Geodesy and Geophysics General Assembly, 13th, Berkeley, California.

Onstott, T. C., and Hargraves, R. B., 1981, Proterozoic transcurrent tectonics: Nature, v. 289, p. 131–136.

Piper, J.D.A., 1982, The Precambrian paleomagnetic record: The case for the Proterozoic Supercontinent: Earth and Planetary Science Letters, v. 59, p. 61–89.

Posadas, V. G., and Kalliokoski, J., 1967, Rb-Sr ages of the Encrucijada granite intrusive into the Imataca Complex, Venezuela: Earth and Planetary Science Letters, v. 2, p. 210–214.

Short, K. C., and Steenken, W. F., 1962, A reconnaissance of the Guayana Shield from Guasipati to the Rio Aro, Venezuela: Association Venezolano de Geologia, Minera, y Petrologia Boletin Informativo, v. 5, p. 189–221.

Steiger, R. H., and Jäger, E., 1977, Subcommission on Geochronology: Convention on the use of decay constants in geo- and cosmochemistry: Earth and Planetary Science Letters, v. 36, p. 359–362.

Van Breemen, O., and Dodson, M. H., 1972, Metamorphic chronology of the Limpopo Belt, southern Africa: Geological Society of America Bulletin, v. 83, p. 2005–2018.

Walraven, F., 1977, The transition between the mafic and the acid phase of the Bushveld Complex in the Rustenburg-Brits area [M.Sc. thesis]: Pretoria, University of Pretoria, 111 p.

Yanase, Y., 1970, Isotopic studies in geological materials—Controlled heating experiments with the $^{40}Ar/^{39}Ar$ dating technique [Ph.D. thesis]: Toronto, Ontario, University of Toronto, 136 p.

York, D., 1969, Least squares fitting of a straight line with correlated errors: Earth and Planetary Science Letters, v. 5, p. 320–324.

York, D., Hall, C. M., Yanase, Y., Hanes, J. A., and Kenyon, W. J., 1981, $^{40}Ar/^{39}Ar$ dating of terrestrial minerals with a continuous laser: Geophysical Research Letters, v. 8, no. 11, p. 1136–1138.

Zijderveld, J.D.A., 1967, A.C. demagnetization of rocks: Analysis of results, *in* Collinson, D. S., Creer, K. M., and Runcorn, S. K., eds., Methods in paleomagnetism: Amsterdam, Elsevier, p. 254–286.

MANUSCRIPT RECEIVED BY THE SOCIETY JUNE 10, 1983
REVISED MANUSCRIPT RECEIVED NOVEMBER 14, 1983
MANUSCRIPT ACCEPTED NOVEMBER 21, 1983

Printed in U.S.A.

Downloaded from gsabulletin.gsapubs.org on March 6, 2010

ERRATA

Volume 95, p. 1053

Figure 9 was published with printer's information on it. The figure as it should have appeared is shown here.

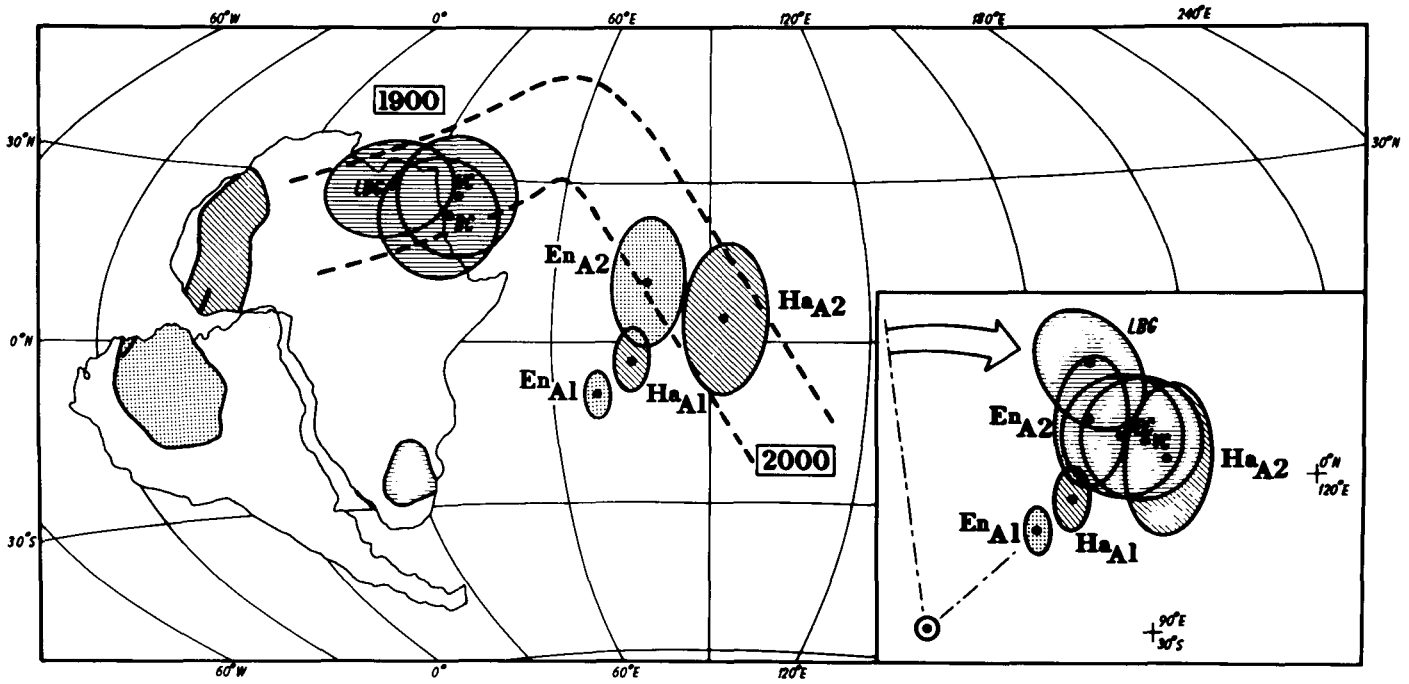


Figure 9. South America and Africa in the Bullard and others (1965) reconstruction. Shown are the Encrucijada and Harper poles (positions and symbols are the same as for Fig. 8), the Kalahari Shield poles and 95% confidence ovals (horizontal hachure), the Bushveld Complex (BC), the Vredefort Dome (VC), and the Limpopo Belt Gneiss (LBG). The dashed apparent polar wander swathe and attached ages (in millions of years) are from Piper (1982). The inset illustrates the excellent juxtaposition of the Kalahari poles on the West African and Guayana poles obtained when the Kalahari Shield is rotated according to the reconstruction of Onstott and Hargraves (1981) (rotation pole 44°E, 27°S, and 64°CW) to align the Limpopo and Ubendian mobile belts. The polarity of poles BC and VC matches the polarity of poles En_{A2} and Ha_{A2}.

Volume 95, p. 1199

Figure 4 was printed incorrectly, and some information, including the scales, was omitted. The correct figure appears below.

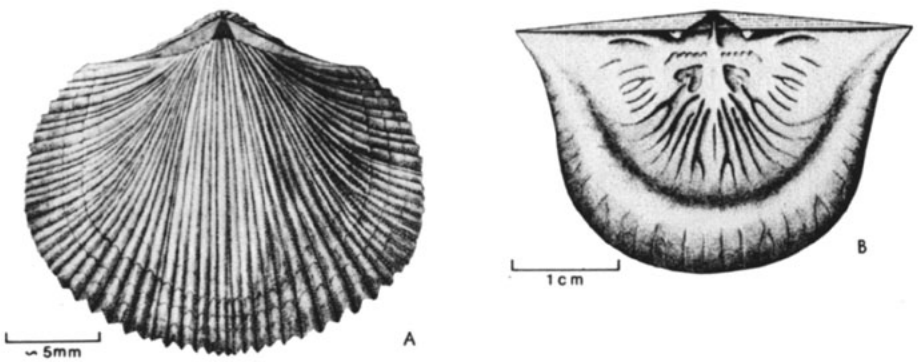


Figure 4. Selected endemic brachiopods from the middle volcanic unit of the Summerford Group, Virgin Arm, New World Island, Newfoundland. A. *Acanthorthis terreneuvensus* Neuman; reconstruction of exterior of dorsal valve based on specimens illustrated by Neuman (1976, Pl. 2, Figs. 22–28) and other paratypes; drawing by Laurie Hughes. B. *Guttasella gutta* Neuman, reconstruction of interior of dorsal valve, from Neuman (1976, p. 32).